



## CATEGORIES

Aluminum Electrolytic Capacitor —

Liquid Small Products

Liquid Large Products

Conductive Polymer Aluminum

Solid Electrolytic Capacitors

Conductive Polymer Hybrid

Aluminum Electrolytic Capacitors

Multilayer Polymer Aluminum

Solid Electrolytic Capacitor

Conductive Tantalum Capacitor

Electric Double Layer Capacitor

Hybrid Supercapacitor

Metallized Polypropylene Film

Capacitors

MLCC

## FEATURED PRODUCTS



LKE

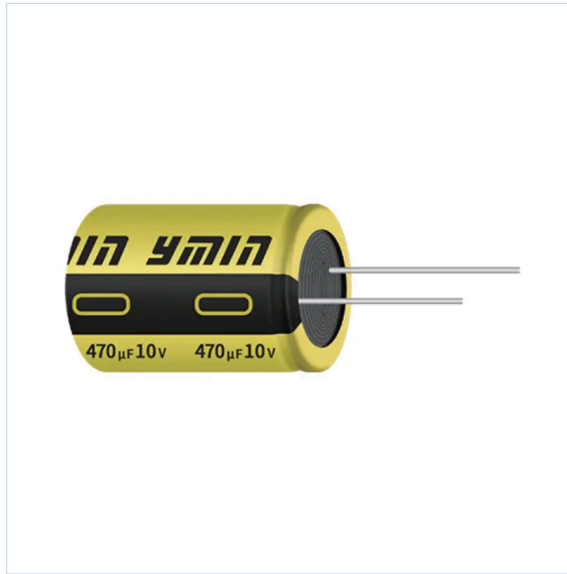


LKD



CW3H

## LKM



### Short Description:

#### Aluminum Electrolytic Capacitor

#### Radial Lead Type

Small size, high frequency and large ripple current resistance,

high frequency and low impedance, power supply products,

7000~10000 hours in 105°C environment, comply with AEC-Q200 RoHS directive.

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PRODUCT DETAIL

LIST OF STANDARD PRODUCTS

PRODUCT TAGS

### Main Technical Parameters

#### Technical Parameter

- ◆ 105°C 7000~10000 Hours
- ◆ Miniature, High Ripple Current, High Frequency, Low ESR
- ◆ RoHS Compliant
- ◆ AEC-Q200 Qualified, Please Consult Us For More Details

#### Specification

Items

Characteristics

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Multilayer ceramic  
chip capacito...

|                                     |                                                                                                                                                                                     |      |                                           |      |      |     |                                                                                        |      |     |  |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------|------|------|-----|----------------------------------------------------------------------------------------|------|-----|--|
| Rated Voltage                       | 10~500V.DC                                                                                                                                                                          |      |                                           |      |      |     | <div>* Email (We will reply you via email in 24 hours)</div>                           |      |     |  |
| Capacitance Tolerance               | ±20%(25±2℃ 120Hz)                                                                                                                                                                   |      |                                           |      |      |     | <div>Where did you hear about our website?</div>                                       |      |     |  |
| Leakage Current((uA)                | 10— 120WV  ≤ 0.01CV or 3uA whichever is greater C:rated capacitance(uF) V:rated voltage                                                                                             |      |                                           |      |      |     | <div>Enter the products number,quantity, and other requirements.</div> <div>Send</div> |      |     |  |
|                                     | 160 ~500WV  ≤ 0.02CV+10(uA) C:rated capacitance(uF) V:rated voltage(V) 2 minutes re                                                                                                 |      |                                           |      |      |     |                                                                                        |      |     |  |
| Dissipation Factor (25±2℃ 120Hz)    | Rated Voltage(V)                                                                                                                                                                    | 10   | 16                                        | 25   | 35   | 50  |                                                                                        |      |     |  |
|                                     | tgδ                                                                                                                                                                                 | 0.19 | 0.16                                      | 0.14 | 0.12 | 0.1 |                                                                                        |      |     |  |
|                                     | Rated Voltage(V)                                                                                                                                                                    | 120  | 160                                       | 200  | 250  | 350 |                                                                                        |      |     |  |
|                                     | tgδ                                                                                                                                                                                 | 0.09 | 0.09                                      | 0.08 | 0.08 | 0.1 | 0.1                                                                                    | 0.12 | 0.2 |  |
|                                     | For those with rated capacitance larger than 1000uF, when the rated capacitance is increased by 1000uF,then tgδ will be increased by 0.02                                           |      |                                           |      |      |     |                                                                                        |      |     |  |
| Temperature Characteristics (120Hz) | Rated Voltage(V)                                                                                                                                                                    | 10   | 16                                        | 25   | 35   | 50  | 63                                                                                     | 80   | 100 |  |
|                                     | Z(-40℃)/Z(20℃)                                                                                                                                                                      | 6    | 4                                         | 3    | 3    | 3   | 3                                                                                      | 3    | 3   |  |
|                                     | Rated Voltage(V)                                                                                                                                                                    | 120  | 160                                       | 200  | 250  | 350 | 400                                                                                    | 450  | 500 |  |
|                                     | Z(-40℃)/Z(20℃)                                                                                                                                                                      | 5    | 5                                         | 5    | 5    | 7   | 7                                                                                      | 7    | 8   |  |
| Endurance                           | After standard test time with applying the rated voltage with the rated ripple current in the oven at 105℃, the following specification shall be satisfied after 16 hours at 25±2℃. |      |                                           |      |      |     |                                                                                        |      |     |  |
|                                     | Capacitance change                                                                                                                                                                  |      | within±20% of the intial value            |      |      |     |                                                                                        |      |     |  |
|                                     | Dissipation Factor                                                                                                                                                                  |      | Not more than 200% of the specified value |      |      |     |                                                                                        |      |     |  |
|                                     | Leakage current                                                                                                                                                                     |      | Not more than the specified value         |      |      |     |                                                                                        |      |     |  |
|                                     | Load life( hours)                                                                                                                                                                   | Φ5   |                                           |      |      |     | 7000hrs                                                                                |      |     |  |
|                                     |                                                                                                                                                                                     | Φ6.3 |                                           |      |      |     | 9000hrs                                                                                |      |     |  |
|                                     |                                                                                                                                                                                     | ≥Φ8  |                                           |      |      |     | 10000hrs                                                                               |      |     |  |
| Shelf Life At High Temperature      | After leaving capacitors under no load at 105℃ for 1000 hours, the following specification shall be satisfied at 25±2℃.                                                             |      |                                           |      |      |     |                                                                                        |      |     |  |
|                                     | Capacitance change                                                                                                                                                                  |      | within±20% of the intial value            |      |      |     |                                                                                        |      |     |  |
|                                     | Dissipation Factor                                                                                                                                                                  |      | Not more than 200% of the specified value |      |      |     |                                                                                        |      |     |  |

\* Name

\* Email (We will reply you via email in 24 hours)

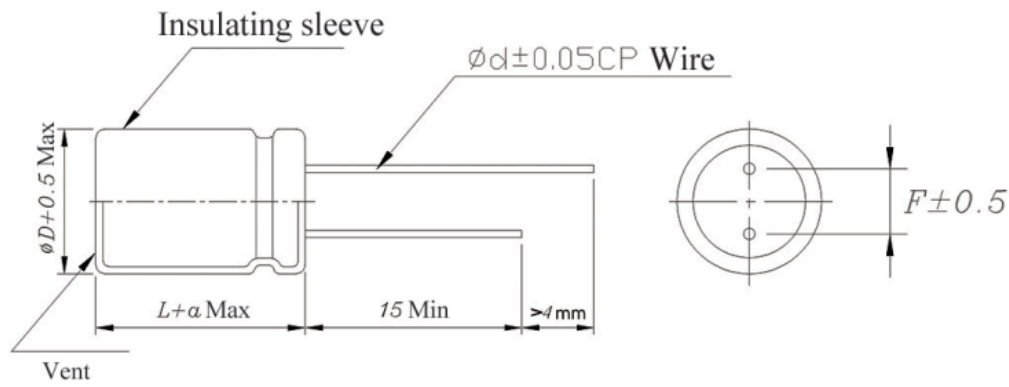
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Product Dimensional Drawing



Remark: capacitors with diameter more than 6.3 have safety vent

Dimension (Unit: mm)

Remark: capacitors with diameter more than 6.3 have safety vent

|      |       |
|------|-------|
| L=9  | a=1.0 |
| L≤16 | a=1.5 |
| L>16 | a=2.0 |

|   |     |     |     |     |      |      |     |     |
|---|-----|-----|-----|-----|------|------|-----|-----|
| D | 5   | 6.3 | 8   | 10  | 12.5 | 14.5 | 16  | 18  |
| d | 0.5 | 0.5 | 0.6 | 0.6 | 0.6  | 0.8  | 0.8 | 0.8 |
| F | 2   | 2.5 | 3.5 | 5   | 5    | 7.5  | 7.5 | 7.5 |

#### Frequency Correction Factor

| Frequency (Hz) | 50  | 120 | 1K  | 10K-50K | 100K |
|----------------|-----|-----|-----|---------|------|
| Coefficient    | 0.4 | 0.5 | 0.8 | 0.9     | 1    |



#### Temperature Correction Factor

| Environment Temperature(°C) | 50°C | 70°C | 85°C | 105°C |
|-----------------------------|------|------|------|-------|
| Correction Factor           | 2.1  | 1.8  | 1.4  | 1     |



#### LKM Series: Small Size, High Energy – Providing Stable Core Power for Modern High-Performance Electronic Devices

In today's era driven by electronic technology, the performance boundaries of electronic devices are constantly being pushed, and their internal designs are becoming increasingly compact and complex. Whether it's speeding electric vehicles, constantly running data servers, or the increasingly compact fast chargers in our hands, all place unprecedentedly stringent demands on one of the most fundamental components – capacitors: they must maintain excellent stability and long-term reliability under extreme temperatures, severe voltage fluctuations, and high-frequency ripple currents. YMIN's LKM series aluminum electrolytic capacitors were born to meet these challenges. With their core characteristics of "small size, high frequency and low impedance, high ripple current, long lifespan, and automotive-grade quality," they have become an indispensable and stable "heart" in many high-end electronic system designs.

##### I. Precise Positioning: A Solution for High Performance and High Reliability

The LKM series is not an ordinary general-purpose aluminum electrolytic capacitor, but a radially led product specifically designed for high-performance applications. Its operating temperature range covers a wide range of -55°C to +105°C, ensuring stable operation, especially at 105°C, with a lifespan of 7000 to 10000 hours. This means that even under severe conditions of internal heat accumulation, LKM capacitors can operate continuously for nearly a year and a half (calculated 24/7) with performance degradation remaining within standard limits. More importantly, the entire series complies with the AEC-Q200 standard, directly signifying that it meets the automotive electronics industry's top requirements for component reliability, environmental adaptability, and long lifespan. From in-vehicle infotainment systems and Advanced Driver Assistance Systems (ADAS) to the electronic control units of new energy vehicles, the LKM series provides the "zero tolerance" failure rate guarantee sought by automotive electronics.

##### II. Core Technological Advantages: Analyzing the Deeper Meaning of "High Performance"

The product description of the LKM series contains several key technological advantages, which together form the cornerstone of its superior performance.

1. Low Impedance and High-Frequency Characteristics: Traditional aluminum electrolytic capacitors experience a significant increase in equivalent series resistance (ESR) at high frequencies, leading to decreased capacitor efficiency and excessive heat generation. The LKM series achieves low impedance characteristics under high frequency conditions through advanced material and process optimization. This means that in high-frequency switching power supplies (such as server power supplies

currents far exceeding those of ordinary capacitors, thanks to its lower internal ESR and optimized heat dissipation design. For example, in the datasheet, a 400V/100 $\mu$ F capacitor can withstand a ripple current of up to 1.8A at 105°C. This characteristic is crucial for applications handling instantaneous high currents, such as inverter circuits in motor drives, uninterruptible power supplies (UPS), and industrial power supplies, ensuring system stability and the longevity of the capacitors themselves under drastic load changes.

3. Superior Lifespan and Stability: The 7000-10000-hour lifespan promised by the LKM series is not an exaggeration, but rather the result of rigorous durability testing. The tests require that after high-temperature full-load testing, the capacitor's capacitance change, loss tangent ( $\tan\delta$ ), and leakage current all remain within the initially specified stringent range. This excellent stability reduces the risk of performance degradation or failure due to capacitor aging, providing crucial quality assurance for equipment requiring continuous operation year-round (such as communication base stations, smart meters, and energy storage systems).

4. Robust Physical Structure and Safety: For capacitors with a diameter greater than 6.3mm, the LKM series is designed with a safety explosion-proof valve. This safety mechanism can release pressure in a controlled manner when the capacitor experiences a surge in internal pressure due to extreme abnormal conditions (such as voltage surges or circuit errors), preventing the casing from rupturing and thus avoiding secondary damage to surrounding circuits, greatly improving the safety of the entire device.

### III. Wide Range of Application Scenarios: Empowering Electronic Innovation Across Industries

The technical characteristics of the LKM series make it shine in several cutting-edge technology fields:

- **Automotive Electronics:** With the wave of automotive electrification and intelligentization, the number of in-vehicle electronic systems has surged. LKM's AEC-Q200 certification makes it an ideal choice for key components such as engine control units (ECUs), battery management systems (BMS), on-board chargers (OBCs), and LED headlight drivers, ensuring flawless operation in harsh environments such as vibration, high and low temperature shocks encountered by vehicles.
- **AI Data Servers and Communication Equipment:** Data center server power supplies require extremely high power density and efficiency. LKM's low ESR and high ripple current capabilities effectively meet the stringent requirements of CPU/GPU power supply modules (VRMs) for decoupling and filtering, ensuring the stable operation of AI computing clusters and cloud computing infrastructure. Similarly, in 5G communication equipment, it can also handle power filtering and signal coupling tasks.
- **Robotics and Industrial Automation:** Servo drives in industrial robots and CNC machine tools need to handle high-frequency, high-current switching operations. LKM capacitors, in the inverters and DC-Link circuits of such devices, can effectively smooth the DC bus voltage, absorb regenerative energy, and improve drive accuracy and system response speed.
- **PD Fast Charging and Smart Lighting:** PD fast charging heads based on gallium nitride (GaN) technology pursue extreme miniaturization and high efficiency. The small size (minimum diameter 5mm) and high-frequency, low-impedance characteristics of LKM perfectly match the high switching frequency requirements of GaN fast charging circuits, making it a key component for achieving high-power, small-size fast charging. In the field of smart LED lighting, it ensures the long lifespan of the driver power supply and stable light output.
- **New Energy and Smart Grid:** In photovoltaic inverters, wind power converters, and smart meters, LKM capacitors play a crucial role in DC-DC conversion, inversion, and filtering. Their long lifespan and high-temperature stability directly impact the power generation efficiency and operation and maintenance costs of the entire energy system.

### IV. Choosing the LKM Series: Choosing Reliability and Superior Performance

In summary, the YMIN LKM series aluminum electrolytic capacitors are not merely simple electronic components; they are high-tech products integrating advanced materials science, precision manufacturing processes, and rigorous quality control. They represent the forefront of modern aluminum electrolytic capacitor technology, designed to address the core pain points faced by engineers designing next-generation high-performance, high-reliability electronic devices.



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